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Polycystic Ovary Syndrome prevalence and associated sociodemographic risk factors: a study among young adults in Delhi NCR, India

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Abstract

Introduction Polycystic Ovary Syndrome (PCOS) is a prevalent yet under-researched endocrinologic disorder affecting females of reproductive age, characterized by menstrual dysfunction, infertility, hirsutism, acne, and obesity. Despite its global prevalence, with rates varying significantly among Asian communities, there is a notable lack of region-specific epidemiological data, particularly for urban areas in India. The aim of this study is to assess the prevalence of PCOS and associated sociodemographic risk factors among young adult females in Delhi and National Capital Region (NCR), India.

Methods This study is comprised of two components: a cross-sectional survey and a systematic review. The cross-sectional survey involved 1,164 college-going females aged 18–25 years in Delhi NCR, with data collected through a structured interview schedule assessing sociodemographic variables and PCOS symptoms. PCOS diagnosis was based on the already diagnosed cases and cases diagnosed during the study (Rotterdam criteria, 2003), through symptoms and additional assessment through ultrasonography. The systematic review analysed prevalence studies from 2010 to 2024 across India, focusing on the similar age group.

Results The study found a high 17.40% prevalence rate of PCOS among the participants, with 70.30% already diagnosed and 29.70% newly diagnosed during the study. The prevalence is significantly higher compared to the pooled prevalence of 8.41% reported in previous studies across India. Sociodemographic factors such as age (20 years and above), higher education, ancestry (East India and immigrants), and nuclear family structure were associated with increased PCOS risk. Conversely, factors like belonging to the OBC category and lower middle class were linked to reduced risk.

Discussion The high prevalence of PCOS in Delhi NCR compared to other regions highlights the need for targeted epidemiological research and intervention strategies in urban settings. The association of PCOS with modern lifestyle factors and socioeconomic status underscores the importance of addressing these determinants in managing PCOS effectively. The study contributes valuable insights into the sociodemographic dimensions of PCOS and calls for more comprehensive studies to inform public health strategies.

Keywords Polycystic Ovary Syndrome (PCOS), Prevalence, Sociodemographic Risk Factors, Young Adults, Delhi NCR

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Introduction

Polycystic Ovary Syndrome (PCOS), described by Stein and Leventhal in 1935 [1], represents one of the most prevalent yet understudied endocrinologic disorders affecting females of reproductive age. Its clinical manifestation as menstrual dysfunction, infertility, hirsutism, acne, and obesity show significant heterogeneity across the lifecycle of women [2]. PCOS afflicted females are more likely to experience infertility, dyslipidaemia, endometrial cancer, vascular disease, both clinical and subclinical, mental problems, decreased glucose tolerance or type 2 diabetic mellitus (T2DM), etc. [3–6]. Reproductive and metabolic abnormalities have also been observed in the offspring of females having PCOS [7]. PCOS being a collection of signs and features, among which no single phenomenon is diagnostically relevant on its own due to the lack of consensus criteria, leading to various challenges. Presently, three sets of diagnostic criteria define PCOS, namely the National Institutes of Health (NIH) criteria (1990), Rotterdam criteria (2003), and Androgen Excess Society criteria (2006), each emphasizing distinct diagnostic markers. The Rotterdam criteria has been found to be more inclusive and is preferred [6].

Globally, PCOS affects approximately 1.55 million women [8], resulting in a significant burden of disability-adjusted life years (DALYs) estimated at 0.43 million [9]. The age-standardized incidence rate of PCOS among women of reproductive age was observed at 82.44 per 100,000 in 2017, marking a 1.45% increase from 2007 [10]. Prevalence rates of PCOS vary widely among Asian communities with a prevalence of 5.60% in China [11], 5.30% in Thailand [12], 15.20% in Iran [13], and 6.30% in Sri Lanka [14]. In comparison to western communities and other Asian communities, Indian women have been reported to have a higher prevalence of PCOS [15]. A prevalence rate of PCOS in Indian women is in the range of 3.70%–36% [16–19]. This high frequency of PCOS, as well as its link with ovulation and menstruation abnormalities, infertility, hair loss, and metabolic issues, cause significant financial burden on the nation [2, 20].

PCOS is rapidly assuming epidemic proportions, particularly in India, which underlines the urgency for comprehensive epidemiological investigations, especially considering the scarcity of region-specific data. Studies show that urban Indian women are more prone to developing PCOS compared to their rural counterparts [21, 22]. This is linked to obesogenic diets and sedentary lifestyles, especially among the urban middle class [23]. These dietary shifts and reduced physical activity contribute to rising obesity and metabolic syndrome in urban India [24], which are pivotal in PCOS aetiology. Concerns about modernization and its effects on health are reflected in the rise of PCOS as a"lifestyle disease"in globalizing India [23]. Consequently, urban areas such as the Delhi National Capital Region (NCR) provide a crucial setting for epidemiological research on PCOS. The region's dense population, diverse socioeconomic strata, and varying levels of healthcare access make it an ideal location for assessing PCOS prevalence and associated risk factors. This study addresses the critical gap in region-specific epidemiological data for urban India, particularly in Delhi NCR, a rapidly modernizing region with unique sociodemographic dynamics. While prior studies highlight rural–urban disparities in PCOS prevalence [21, 22], few focus on metropolitan youth, a population increasingly vulnerable to lifestyle-driven endocrine disorders. Our findings provide actionable insights for public health strategies tailored to urban settings, complementing national efforts to mitigate PCOS-related morbidity.

Furthermore, focusing on the age group (18–25 years) enables early detection and intervention, crucial for mitigating long-term health complications associated with PCOS. Therefore, the present study aims to assess prevalence of PCOS among young adult females in the Delhi NCR. To facilitate the relevance of the above data, a systematic review was also conducted to assess the prevalence of PCOS in India from 2010 to 2024 particularly in age group 18–25 years ± 5 years. Additionally, the present study also aims to study the association of various sociodemographic risk factors for PCOS among young adult females in the selected region.

Material and Methods

The present study comprises of two components: a cross-sectional study for the prevalence and associated sociodemographic risk factors among young adult females residing in Delhi NCR and a systematic review for the prevalence of PCOS in India from 2010 to 2024 particularly in age group 18–25 years ± 5 years.

For the prevalence and associated sociodemographic risk factors

Study area and collection of data

The present study is a stratified random cross-sectional study conducted among 1,164 college-going young adult females aged 18 to 25 years (mean age 19.8 ± 1.9 years) residing in Delhi-NCR. The supplementary Table 1 represents the number of individuals whose data was collected and screened for PCOS during fieldwork from different institutions. Data collection was performed through face-to-face interaction. A pre-tested interview schedule was used to collect personal data including socio demographic (age, religion, category, ancestry, education, occupation, marital status, family structure, mothers and fathers' education, and socio-economic status), and reproductive variables (related to PCOS screening).

Measures

Socioeconomic status (SES) was estimated using the Modified Kuppuswamy Scale, which evaluates SES as a combination of education, income, and occupation of a family and generates a numerical score on a scale of 0–29 [25]. Scores of 10 or below were categorized as the lower class, 11–15 as the lower middle class, 16–25 as the upper middle class, and 26–29 as the upper class [25].

This study aligns with contemporary recommendations, including those from Azziz et al. (2019), to enhance comparability and methodological rigor in PCOS epidemiology research [26]. For identifying PCOS cases and non PCOS individuals a pretested interview schedule was used, it comprised of questions related to menstrual cycle (for ovulatory dysfunction), and Hirsutism (for hyperandrogenism). Ovulatory dysfunction refers to oligomenorrhea (disturbed menstruation having cycles more than 35 days apart but less than six months apart) or amenorrhea (absence of menstruation for six to 12 months after a cyclic pattern has been established) [26, 27]. Hirsutism was defined by the presence of a modified Ferriman-Gallwey score of 8 or higher based on participants self-report. The Ferriman-Gallwey score was determined as the cumulative score of presence of terminal hair on the nine body parts, including the upper lip, chin, chest, upper and lower abdomen, thighs, upper and lower back, and upper arms which were graded in range of 0–4 [28]. Participants reported hair growth patterns that persist despite depilation, which was carefully considered during assessment.

Selection of PCOS cases and Non PCOS

Cases were identified in the following manner, based on the interview schedules:

1. Females who self-report being previously diagnosed with PCOS by a physician, meeting the inclusion and exclusion criteria were considered PCOS cases.
2. Females who have two symptoms as per Rotterdam criteria (2003) were considered confirmed cases and were further taken for physician evaluation. The Rotterdam criteria define PCOS by the presence of a minimum of two features out of three (ovulatory dysfunction, polycystic ovaries, and clinical hyperandrogenism) [29].
3. Later, females with one symptom either hyperandrogenism or ovulatory dysfunction were classified as Probable PCOS cases, which were further examined for polycystic ovaries with the help of ultrasonographic assessment. Polycystic ovary morphology (PCOM) was defined as the presence of more than 12 peripheral follicles on transabdominal ultrasound, each measuring 2 to 8 mm in size, together with echogenic ovarian stroma and/or increased ovarian volume (> 10 cm3) [30].

To further ensure diagnostic accuracy, newly diagnosed cases (29.70%) underwent clinical evaluation, including ultrasonography and medical record reviews, to exclude conditions mimicking PCOS (e.g., thyroid dysfunction, hyperprolactinemia). This two-step screening process (self-reporting + clinical evaluation) aligns with the Rotterdam (2003) criteria and minimizes the risk of misclassification.

This confirmed the PCOS cases for the present study. PCOS cases were classified into four phenotypes based on the Rotterdam criteria [26]: Phenotype A: Hyperandrogenism, ovulatory dysfunction, and polycystic ovarian morphology. Phenotype B: Hyperandrogenism and ovulatory dysfunction. Phenotype C: Ovulatory dysfunction and polycystic ovarian morphology and Phenotype D: Hyperandrogenism and polycystic ovarian morphology.

Exclusion criteria for cases: Women who self-reported of being diagnosed by a physician with a history of thyroid abnormalities, corticosteroid consumption, Cushing's syndrome, congenital adrenal hyperplasia, androgen secreting tumours, pregnant or lactating mothers were excluded from the study. In addition, women who had received any hormone treatment or insulin-lowering agent during the preceding 3 months were excluded. This approach, while limiting the representation of these groups, was adopted to ensure accurate assessment of PCOS prevalence. These exclusions are discussed further in the limitations.

Non PCOS: Females included in this group were age-matched and categorized based on the following criteria: those with no features of PCOS (n = 901) and those with one feature, such as regular menstrual cycles or clinical hyperandrogenism (n = 61). Additionally, none of the participants had any major illnesses, and all reported not taking medications or oral contraceptive pills in the past 6 months. Data from non-PCOS participants were used to establish baseline normative ranges for health characteristics, providing a reference point for comparison with PCOS cases [26]. All participants were residing in Delhi NCR.

The study protocol was approved by the Departmental Ethics Committee, Department of Anthropology, University of Delhi (Approval number: Ref. No. anth/2022–23/16). Informed written consent, typed in English and local language, was obtained from each participant prior to their recruitment.

Statistical analyses

Statistical analyses were performed using SPSS version 22 (IBM –SPSS Inc. Chicago, IL) and MS Excel 19. Pair-wise deletion was used for missing data. Categorial variables have been presented as frequencies along with respective percentages, and continuous variables as means along with respective standard deviations (SD). Differences in the distribution of categorical variables were ascertained using the chi-square test. Logistic regression analyses were performed to calculate the odds ratio for understanding the association of the sociodemographic risk factors with PCOS. A p-value <0.05 was considered statistically significant.

Methodology used for systematic review for the prevalence of PCOS in India from 2010 to 2024 particularly in age group 18–25 years ± 5 years

The study framework was designed as per the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines before starting the literature search. Figure 1 depicts the PRISMA flow diagram showing the steps in the systematic review of the study. No adjustments were made after that. The aim and objective of the study were to conduct a systematic review to assess the prevalence of PCOS in India from 2010 to 2024 using NIH, Rotterdam, and AE-PCOS Society criteria particularly in age group 18–25 years ±5 years. This approach accommodates variability in Indian studies, which often group adolescents and young adults together, and aligns with global diagnostic frameworks such as the Rotterdam criteria, ensuring a comprehensive comparison with our primary study focused on college-going females aged 18–25 years. A systemic computer-assisted literature search of all major databases including PubMed, Scopus, and Google Scholar. The following search terms were entered as medical subject headings for finding studies reporting the prevalence of PCOS: The search strategy used a combination of different codes ("polycystic ovary syndrome"OR PCOS) OR "Stein Leventhal Syndrome" OR "polycystic ovarian disease"), (prevalence OR proportion) 'ovary polycystic disease'/exp OR 'ovary polycystic disease' AND (prevalence OR epidemiology) AND ("reproductive aged women"OR"reproductive age") AND India AND"cross sectional study". References in the identified studies were also investigated to identify additional studies.

Eligibility criteria

Inclusion criteria Studies meeting the following criteria were included: (1) cross-sectional, case–control, or cohort studies including PCOS women aged 18–25 years ±5 years years and age-matched controls of any ethnicity; (2) PCOS was diagnosed based on Rotterdam, or self-reported diagnosed by physician; (3) studies containing original data (independent of other studies); (4) design where the prevalence of PCOS with sample size was presented; and (5) publications in full text written in English.

Exclusion Criteria The studies were excluded, if these (1) contained data overlapping data with other studies (2) reported in language other than English (3)

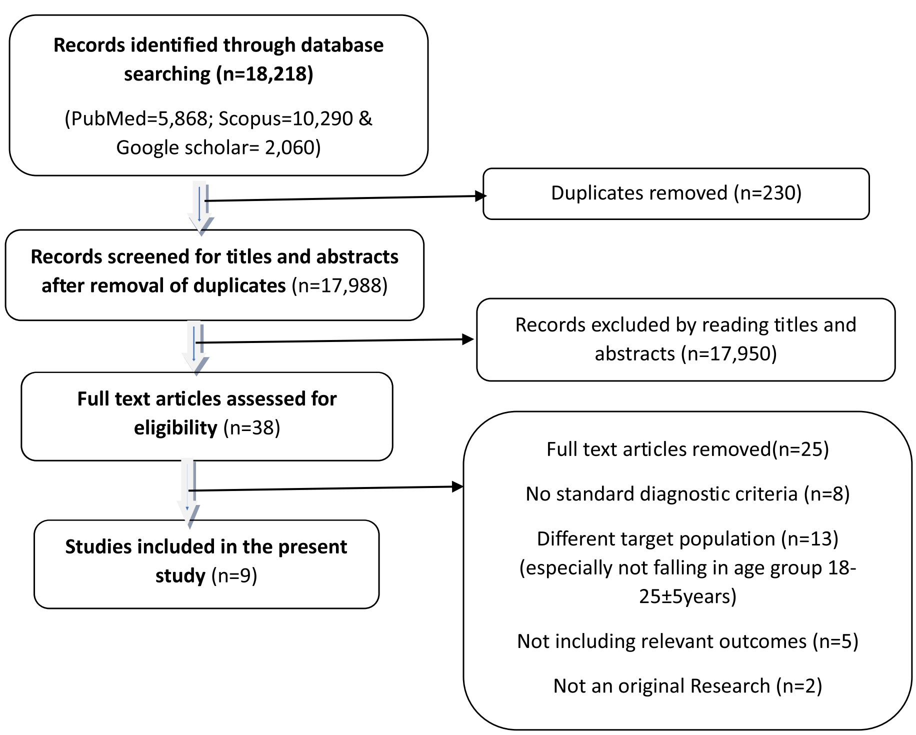

Fig. 1 PRISMA flow diagram showing the steps in the systematic review of the study

epidemiological studies reporting prevalence in family members of affected cases (4) letters, abstracts, and conference proceedings, which are not fully published in peer reviewed journals or published with limited access.

Data extraction

A data extraction form consists of information needed for the study (name of first author, year of publication, country, study design, study population size and description, age group, diagnostic criteria used, and prevalence rates).

Pooled prevalence

To calculate pooled prevalence the prevalence of each study was multiplied by its weight, then the weighted prevalences were summed up.

Results

The present study involves, a total of 1164 participants (age 18–25 years) with mean age of 19.80 ±1.90 S.D. The general characteristics of the sample are presented in Table S2. Of 1,164 participants, 54.80% were below 20 years of age, 83.30% were undergraduates, 67.30% were from the (upper + lower) middle class, 84.50% followed Hinduism, and 64.90% were from General social category.

The overall prevalence of PCOS is 17.40% (202 individuals) (Table 1) Out of the total 202 PCOS cases 70.30% (142; 12.20% of total sample) were already diagnosed with PCOS and 29.70% (60; 5.20% of total sample) were diagnosed during the present study. Out of the 60 participants diagnosed during the present study, the distribution of phenotypes was as follows: Phenotype A (6.50% of PCOS cases), Phenotype B (12.40% of PCOS cases), and Phenotype C (10.80% of PCOS cases). None of the participants exhibited Phenotype D which is a combination of polycystic ovaries and hyperandrogenism symptoms, which may be due to the absence of reporting of biochemical symptoms of hyperandrogenism.

Flow chart is shown in Fig. 2 which depicts study stages for recruitment of PCOS cases and non PCOS group.

The prevalence of PCOS in the present study conducted in Delhi NCR in 2024 is 17.40% among college-going women aged 18–25 years (Table 2). The pooled prevalence of previous studies in the similar age group across various regions in India is 8.41%. Gill et al. (2012) reported a prevalence of 3.70% in Lucknow among 1520 females aged 18–25 years, and Joshi et al. (2014) found a prevalence of 22.50% in Mumbai among adolescents and young adolescents (15–24 years) as per the Rotterdam diagnostic criteria [18, 31]. Other studies, such as Vijaya and Bharadwaj (2014) in Pondicherry and Gupta et al. (2018) in Madhya Pradesh, reported prevalence rates of 11.76% and 8.20%, respectively [32, 34].

The present study reveals that PCOS prevalence is significantly higher in females aged 20 years and above (23.40%) (Table 3) and were Post graduate and above educated. The highest PCOS prevalence among the categories were found in Scheduled Tribes (21.40%) and General/UR (19.90%) categories compared to OBC and SC. Females with higher education levels (24.20%) and having grandparents/parents from outside India (35%), as well as those in nuclear families (21.80%), have a significant higher prevalence of PCOS. Higher mother's education, females with illiterate fathers and upper-class socio-economic status have higher prevalence of PCOS. Additionally, females from Sikh, Christian, and Muslim backgrounds have a higher prevalence of PCOS, though not significant.

Odds ratios (O.R.) analysis was done to understand the association of sociodemographic variables with PCOS (Table 4). O.R. was adjusted for all the variables found significant in the distribution table. Odds ratio analysis indicates that higher age (above 20 years); immigrants (females whose ancestry is from outside India) and females from east India; and females living in nuclear family pose significant increased risk for PCOS compared to the reference categories. OBC category and lower middle class poses a significantly

Table 1 Prevalence of Polycystic Ovary Syndrome in the studied population

		N = 1164	%
No PCOS		962	82.60
PCOS Confirmed		202	17.40
PCOS cases Confirmed (N = 202)			
Already confirmed (70.30%)		142	70.30
Confirmed during the study using Rotterdam Criteria 60 (29.70%)	Phenotype A (Ovary dysfunction + Polycystic ovaries + Hyperandrogenism)	13	6.50
	Phenotype B (Hyperandrogenism + Ovary dysfunction)	25	12.40
	Phenotype C (Ovary dysfunction + Polycystic ovaries)	22	10.80

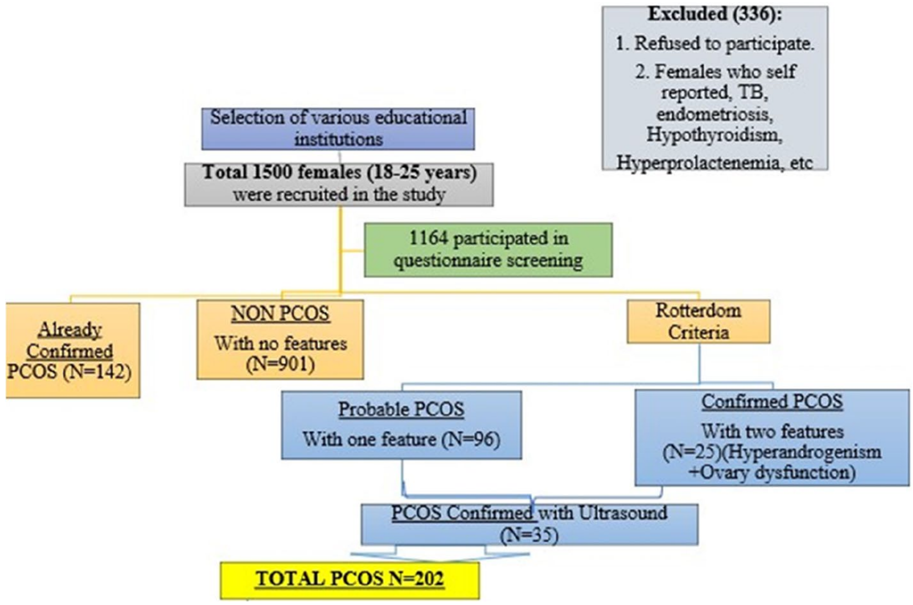

Fig. 2 Flow chart showing study stages for recruitment of PCOS cases and non PCOS group

Table 2 Showing detailed characteristics of the included studies for the review

Author	Year of publication	Prevalence (%)	Study area	Criteria used for prevalence	Sample size	Age group (years)
Gill et.al. [31]	2012	3.70	Lucknow	Rotterdam	1520	18–25
Joshi et.al. [18]	2014	22.50	Mumbai	Rotterdam	600	15–24 (Adolescents and young adolescents Community based)
Vijaya & Bharadwaj [32]	2014	11.76	Pondicherry	Rotterdam	238	19–25
Bharathi et.al. [8]	2017	6	Chennai	Rotterdam	566	18–24
Joseph et.al. [33]	2016	9.10	Karnataka	modified version of Cronin questionnaire	441	20.4 ± 1.5 (Medical and dental college students)
Gupta et.al. [34]	2018	8.20	Madhya Pradesh	Rotterdam	500	17–24
Nanjaiah & Roopadevi [35]	2018	4.55	Karnataka	Rotterdam	396	18–30
Jabeen et.al. [36]	2022	6.80	Telangana	already diagnosed	250	15–25
Shringarpure et.al. [37]	2023	12	Gujarat	Rotterdam	308	19.4 ± 1.7 (medical students)
	Pooled Prevalence	**8.41% (405)**			4819	
Present study	**2024**	**17.40% (202)**	**Delhi NCR**	**Rotterdam**	**1164**	**18–25 years (College going)**

PCOS polycystic ovarian syndrome

reduced risk of PCOS as compared to UR category and upper class, respectively. Females with illiterate mothers have a reduced risk of PCOS, while those with illiterate fathers have a 5.6-fold increased risk.

Discussion

Studies across India from 2010–2024 reported PCOS prevalence ranging from as low as 3.70 percent in a study by Gill et.al., (2012) in Lucknow among 1520 females

Table 3 Distribution of Sociodemographic variables among females with and without PCOS

		Non PCOS Count (Row N%)	PCOS Count (Row N%)	Sig
Age (years)	Below 20	559(87.60)	79(12.40)	0.0**00***
	20 & above	403(76.60)	123(23.40)	
Religion	Hindu	826(83.90)	158(16.10)	0.063
	Muslim	85(76.60)	26(23.40)	
	Christian	24(75.00)	8(25.00)	
	Sikh	9(64.30)	5(35.70)	
	Jain/Buddhist/Not disclosed	18(78.20)	5(21.80)	
Category	General (UR)	544(80.10)	135(19.90)	**0.013***
	Other Backward Class (OBC)	202(88.20)	27(11.80)	
	Scheduled Caste (SC)	98(88.30)	13(11.70)	
	Scheduled Tribe (ST)	22(78.60)	6(21.40)	
Occupation	Student	957(82.60)	201(17.40)	0.687
	Employed	3(75.00)	1(25.00)	
Education	Undergraduate (UG)	815(84.00)	155(16.00)	**0.005***
	Post Graduate and above	147(75.80)	47(24.20)	
Region from where parents or grand-parents originated	North	628(84.00)	120(16.00)	**0.024***
	Central	22(91.70)	2(8.30)	
	South	47(77.00)	14(23.00)	
	West	4(100.00)	0(0.00)	
	East	151(79.90)	38(20.10)	
	Northeast	30(76.90)	9(23.10)	
	Immigrants	26(65.00)	14(35.00)	
Family Type	Joint	187(83.90)	52(16.10)	**0.043***
	Nuclear	697(78.20)	134(21.80)	
Mother Education	UG and above	392(77.30)	115(22.70)	**0.000***
	School level education	380(87.90)	52(12.10)	
	Illiterate	57(86.40)	9(13.60)	
Father Education	UG and above	455(79.60)	116(20.40)	**0.01***
	School level education	339(86.70)	52(13.30)	
	Illiterate	17(73.90)	6(26.10)	
Socio Economic Status	Upper class	213(75.80)	68(24.20)	**0.000***
	Upper Middle	348(80.40)	85(19.60)	
	Lower Middle	186(91.60)	17(8.40)	
	Lower	125(86.20)	20(13.80)	

UR Unreserved

*Significance is shown by *p* value is < 0.05

aged 18–25 years, to as high as 22.50 percent in a study conducted by Joshi et.al., (2014) in a community-based settings in Mumbai among 600 girls (15–24 years) [18, 31]. Literature suggests an increase in PCOS prevalence over the last decade [38], which can also be observed in the systematic review where the prevalence of PCOS is reported to be 3.70 percent in 2012 in Lucknow to 12 percent in 2023 in Gujarat [31, 37]. As per the nine studies in the review, pooled prevalence of PCOS is 8.41 percent among young adults across India. This is lower (almost half) than the 17.40 percent prevalence found in the present cross-sectional study conducted in Delhi NCR, with almost all studies except Joshi et al.'s (2014) study in Mumbai reporting higher prevalence (22.50 percent) [18]. This is higher than the global prevalence of 10.89 percent according to a systematic review and meta-analysis revealed [39]. Delhi NCR and Mumbai, being metropolitan cities with heterogenous populations and many people come from outside primarily for work and academic opportunities [40]. This displacement causes psychological stress, disrupted sleep routine, and poor dietary habits resulting from competition for education

Table 4 Regression analysis for association of sociodemographic variables with PCOS

		O.R	95% Confidence Interval for Exp(B)		Sig
			Lower Bound	Upper Bound	
Age	Below 20	-	-	-	
	20 & above	1.752	1.149	2.672	**0.009***
Category	General (UR)	-	-	-	
	Other Backward Class (OBC)	0.533	0.304	0.936	**0.028***
	Scheduled Caste (SC)	0.675	0.313	1.456	0.317
	Scheduled Tribe (ST)	0.647	0.172	2.426	0.518
Education	Undergraduate (UG)	-	-	-	
	Post Graduate and above	0.964	0.501	1.856	0.913
Ancestry	North	-	-	-	
	Central	0.631	0.139	2.864	0.551
	South	1.702	0.725	3.994	0.222
	West	0.000	0.000		0.999
	East	1.771	1.113	2.820	**0.016***
	Northeast	1.282	0.438	3.754	0.650
	Immigrants	2.464	1.162	5.224	**0.019***
Family Type	Joint	-	-	-	
	Nuclear	1.676	1.094	2.569	**0.018***
Mother Education	UG and above	-	-	-	
	School level education	0.433	0.165	1.135	0.089
	Illiterate	0.434	0.263	0.717	**0.001***
Father Education	UG and above	-	-	-	
	School level education	1.445	0.826	2.527	0.197
	Illiterate	5.592	1.373	22.774	**0.016***
Socio Economic Status	Upper class	-	-	-	
	Upper middle	0.893	0.584	1.366	0.601
	Lower middle	0.414	0.203	0.842	**0.015***
	Lower	0.708	0.317	1.583	0.401

*Significance is shown by p value is < 0.05. '-' indicates Reference category. O.R. is adjusted for age, education, category, ancestry, family type, mother's and father's education and socioeconomic status

and jobs, and juggling responsibilities of home and work, that could possibly be associated with rising rates of PCOS [23, 41–43].

We have observed that females from this upper and upper middle class are at increased risk for PCOS in the present study. In concordance to the present study media accounts have suggested that PCOS is on the rise in India and most prevalent among the urban middle and upper classes because of their lifestyles [44–46]. The features of the "modern" upper and middle-class lifestyles associated with PCOS is relatively recent in India [23]. In 1991, economic liberalization marked a watershed moment, leading to rapid economic growth and the emergence of new upper and middle classes [47]. Higher disposable incomes have contributed to a "nutrition transition" [48], with increased consumption of fats, oils, processed foods, mixed carbohydrate diets, and sugars [49–51]. Eating out has become integral to India's urban-centred public culture and a marker of higher socioeconomic status [52–54]. Increased access to labour-saving devices and cars has reduced physical activity, leading to rising prevalence of insulin resistance and obesity, which are all associated with PCOS pathogenesis [24, 55–57]. Consequently, PCOS may result from a mismatch between modern environmental conditions and the evolutionary adaptations of our ancestors [58–60].

This can also be understood through the thrifty gene hypothesis, suggesting genes that were advantageous for survival in resource-scarce environments are now contributing to higher PCOS rates in resource-abundant urban settings [60, 61]. Neel (1962) proposed this hypothesis, attributing the onset of type-II diabetes to the human gene pool's thrifty nature [62]. This genetic trait was advantageous during the hunting and gathering era when food was scarce, helping humans survive periods of famine. However, in modern society with abundant food, this thrifty genotype leads to fat accumulation in the body, which remains unused and results in lifestyle diseases like PCOS [61, 62]. The present study also revealed that higher age (20 years and above) is associated with PCOS. This trend aligns with findings from other studies, where the prevalence of PCOS increases with advancing age, peaking around the 20–24 age group [2, 63, 64]. Literature suggests that clinical presentations of PCOS vary with age; women in the age group 20–29 years tend to exhibit higher levels of androgens and more pronounced hyperandrogenism symptoms [65] which often leads to other severe health conditions Therefore, early diagnosis and lifestyle modifications are crucial for managing PCOS and preventing long-term reproductive and metabolic complications [63].

Furthermore, this is the first study to examine PCOS across various caste categories. The results reveal that the Other Backward Classes (OBC) category has a lower risk of PCOS compared to the General category. Given that PCOS is the most prevalent endocrine and metabolic disorder in adult women, its phenotypic expression can vary significantly across ethnicities [66–69]. This finding may reflect broader societal disparities and variations in access to healthcare and resources among caste groups [70, 71], underscoring the need for targeted interventions and deeper exploration into how caste intersects with health outcomes in PCOS.

The present study also highlights a significant relationship between PCOS and the place of origin of parents/grandparents, particularly among females with roots outside of India, which comprises mainly those hailing from bordering regions of India, i.e., Pakistan, Bangladesh, and Nepal. PCOS prevalence in these countries varies, with studies reporting rates ranging from 10.90 percent among female medical students in Karachi [72] to as high as 92.16 percent among those seeking consultation for hirsutism in Bangladesh [73]. Similarly, a study among medical students in Bangladesh reported a prevalence of 37 percent [74], while in Nepal, approximately 5–10 percent of the total population is affected by PCOS [75]. According to the regional distribution of PCOS in India, the present study highlights a notably higher risk among East Indian women (20.10 percent). In concordance to the present study, a study by Moghul (2015), reported an alarming 25.88 percent prevalence of PCOS in East India [76]. Immigrants show a higher prevalence of PCOS in the present study, this is well expected as the immigrants in the present study are from Bangladesh, Pakistan, and East Indian regions where the prevalence of PCOS in general is reported to be high [72–76]. This, coupled with the psychological stresses associated with displacement, further can exacerbate the PCOS condition, resulting in higher PCOS prevalence in individuals in India when migrated from their respective regions of origin.

Moreover, females living in nuclear families demonstrate higher risks of PCOS compared to those in joint families. In recent years, the traditional structure of joint family system has experienced remarkable changes due to the impact of modernization, globalization, and the changing aspirations of the Indian population [77]. Living in a nuclear family may lead to increased stress [78, 79], sedentary lifestyle, or obesogenic dietary patterns [80] that contribute to PCOS. In joint families, social support and shared responsibilities may mitigate some risk factors [81]. This suggests potential psychosocial or environmental influences on PCOS development within different family contexts.

There have not been many studies which report the association of parental education on PCOS. In the present study, females whose mothers are illiterate (compared to those with PG and higher education) may have a reduced risk of PCOS, potentially due to illiteracy correlating with lower socioeconomic status or lifestyle factors [82] that are protective against PCOS. Females whose fathers are illiterate (compared to those with higher education) pose an increased risk for PCOS. Lower paternal education level influences the household socioeconomic status, further access to healthcare, or lifestyle factors that affect PCOS risk [83]. These findings highlight the complex interplay between socio demographic factors, ethnicity, and PCOS. These findings have important implications for healthcare planning and clinical practice.

Strengths

This study integrates a large cross-sectional survey with a systematic review to provide a robust epidemiological perspective on PCOS, using the Rotterdam criteria for accurate diagnosis and focusing on sociodemographic risk factors to enhance understanding of PCOS in urban settings. By emphasizing early detection among young adults (18–25 years), it offers critical insights for timely intervention and public health planning.

Limitations

This study has certain limitations that should be acknowledged. First, the exclusion of conditions mimicking PCOS (e.g., hypothyroidism, hyperprolactinemia, late-onset congenital adrenal hyperplasia) relied on participants' self-reports of physician-diagnosed conditions, which may introduce bias. Second, the absence of hormonal assays limited the identification of biochemical hyperandrogenemia and certain PCOS phenotypes. Third, excluding participants on hormonal medications, pregnant or lactating women, and those with a history of surgery may have led to underrepresentation of PCOS cases. Future research should analyse these subgroups separately to enhance understanding of PCOS prevalence and risk factors. Lastly, the study's cross-sectional design and sample population, while diverse, may not fully represent all young adults in Delhi NCR or establish causal relationships. Our focus on college-going females may limit generalizability to non-college or rural populations, who may experience distinct sociodemographic and lifestyle risk factors.

Future studies addressing these limitations can improve the robustness and applicability of PCOS research for public health interventions.

Conclusion

In conclusion, the prevalence of PCOS is high among young adult females residing in Delhi NCR which is second highest among the studies conducted in India in the similar age group. Additionally, sociodemographic factors such as higher age, ancestry from outside and east India, parental education, nuclear family type, and higher socioeconomic status were found to be associated with higher PCOS prevalence, this suggests potential genetic, psychosocial, or environmental influences. Together, all these risk factors synergize to create a perfect storm of health pressures implicated in PCOS. They also amplify the long-term risks of PCOS such as the risk of diabetes, cardiometabolic risk factors and the cumulative effect of exposure to these stressors can affect the illness trajectory of PCOS, worsening outcomes. Therefore, the present study highlights the need for targeted interventions tailored to specific demographic groups to mitigate PCOS-related complications for improved health outcomes and healthcare planning. Public health interventions which can raise awareness and encourage early diagnosis and lifestyle changes, while improving healthcare accessibility and education about PCOS is crucial.

Abbreviations

PCOS	Polycystic Ovary Syndrome
PCOM	Polycystic Ovary Morphology
T2DM	Type 2 Diabetic Mellitus
NIH	National Institutes of Health
DALYs	Disability-Adjusted Life Years
NCR	National Capital Region
SES	Socioeconomic Status
SPSS	Statistical Package for the Social Sciences
SD	Standard Deviation
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
AE-PCOS	Androgen Excess Polycystic Ovary Syndrome
IL	Illinois (as in Chicago, IL, for location)
UG	Undergraduate
PG	Post Graduate
O.R.	Odds Ratio
UR	Unreserved (General category)
OBC	Other Backward Class
SC	Scheduled Caste
ST	Scheduled Tribe

Supplementary Information

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Supplementary Material 1.

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Declaration of generative AI in scientific writing

Authors did not use generative artificial intelligence (AI) and AI-assisted technologies in the writing process or to improve readability and language of the manuscript.

Authors' contributions

Conceptualisation of the manuscript was done by AS and KNS. Main writing, data collection and analysis was done by AS. Resources were provided by KNS and YS. KNS, YS, and NKD reviewed the manuscript.

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Data availability

Data can be available after special request from the corresponding author, due to participants privacy concerns.

Declarations

Ethics approval and consent to participate

The study protocol was approved by the Departmental Ethics Committee, Department of Anthropology, University of Delhi (Approval number: Ref. No. anth/2022–23/16). Informed written consent, typed in English and Hindi language, was obtained from each participant prior to their recruitment.

Competing interests

The authors declare no competing interests.

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References

1. El Hayek S, Bitar L, Hamdar LH, Mirza FG, Daoud G. Poly Cystic Ovarian Syndrome: An updated overview. Front Physiol. 2016;7:124. https://doi.org/10.3389/fphys.2016.00124.
2. Motlagh Asghari K, Nejadghaderi SA, Alizadeh M, Sanaie S, Sullman MJM, Kolahi A-A, et al. Burden of polycystic ovary syndrome in the Middle East and North Africa region, 1990–2019. Sci Rep. 2022;12:7039. https://doi.org/10.1038/s41598-022-11006-0.
3. Lim SS, Davies MJ, Norman RJ, Moran LJ. Overweight, obesity and central obesity in women with polycystic ovary syndrome: a systematic review and meta-analysis. Hum Reprod Update. 2012;18(6):618–37.
4. Wild RA. Dyslipidemia in PCOS. Steroids. 2012;77(4):295–9.
5. Peigné M, Dewailly D. Long term complications of polycystic ovary syndrome (PCOS). Ann Endocrinol (Paris). 2014;75(4):194–9.
6. Teede HJ, Misso ML, Costello MF, et al. Recommendations from the international evidence-based guideline for the assessment and management of polycystic ovary syndrome. Hum Reprod. 2018;33(9):1602–18.
7. Yu HF, Chen HS, Rao DP, Gong J. Association between polycystic ovary syndrome and the risk of pregnancy complications: A PRISMA-compliant

systematic review and meta-analysis. Medicine (Baltimore). 2016;95(51). https://doi.org/10.1097/MD.0000000000004863
8. Vidya Bharathi R, Swetha S, Neerajaa J, Varsha Madhavica J, Janani DM, Rekha SN, et al. An epidemiological survey: Effect of predisposing factors for PCOS in Indian urban and rural population. Middle East Fertil Soc J. 2017;22(4):313–6. https://doi.org/10.1016/j.mefs.2017.05.007.
9. Singh S, Pal N, Shubham S, Sarma DK, Verma V, Marotta F, et al. Polycystic ovary syndrome: Etiology, current management, and future therapeutics. J Clin Med. 2023;12(4):1454.
10. Liu Y, Li J, Yan Z, Liu D, Ma J, Tong N. Improvement of insulin sensitivity increases pregnancy rate in infertile PCOS women: A systemic review.
11. Li R, Zhang Q, Yang D, Li S, Lu S, Wu X, et al. Prevalence of polycystic ovary syndrome in women in China: a large community-based study. Hum Reprod. 2013;28(9):2562–9.
12. Kaewnin J, Vallibhakara O, Vallibhakara SA, Wattanakrai P, Butsripoom B, Somsook E. Prevalence of polycystic ovary syndrome in Thai University adolescents. Gynecol Endocrinol. 2017;33(8):653–6.
13. Mehrabian F, Khani B, Kelishadi R, Ghanbari E. The prevalence of polycystic ovary syndrome in Iranian women based on different diagnostic criteria. Endokrynol Pol. 2011;62(3):238–42.
14. Kumarapeli V, Seneviratne Rde A, Wijeyaratne CN, Yapa RMS, Dodampahala SH. A simple screening approach for assessing community prevalence and phenotype of polycystic ovary syndrome in a semi-urban population in Sri Lanka. Am J Epidemiol. 2008;168(3):321–8.
15. Rao VS, Cowan S, Armour M, Smith CA, Cheema BS, Moran L, et al. A global survey of ethnic Indian women living with polycystic ovary syndrome: Co-morbidities, concerns, diagnosis experiences, quality of life, and use of treatment methods. Int J Environ Res Public Health. 2022;19(23):15850.
16. Nidhi R, Padmalatha V, Nagarathna R, Amritanshu R. Prevalence of polycystic ovarian syndrome in Indian adolescents. J Pediatr Adolesc Gynecol. 2011;24(4):223–7.
17. Nair MKC, Pappachan P, Balakrishnan S, Leena ML, George B, Russell PS. Menstrual irregularity and polycystic ovarian syndrome among adolescent girls--a 2 year follow-up study. Indian J Pediatr. 2012;79(S1)
18. Joshi B, Mukherjee S, Patil A, Purandare A, Chauhan S, Vaidya R. A cross-sectional study of polycystic ovarian syndrome among adolescent and young girls in Mumbai. India Indian J Endocrinol Metab. 2014;18(3):317–24.
19. Ganie MA, Hassan S, Nisar S, Shamas N, Rashid A, Ahmed I, et al. High-sensitivity C-reactive protein (hs-CRP) levels and its relationship with components of polycystic ovary syndrome in Indian adolescent women with polycystic ovary syndrome (PCOS). Gynecol Endocrinol. 2014;30(11):781–4.
20. Azziz R, Marin C, Hoq L, Badamgarav E, Song P. Health care-related economic burden of the polycystic ovary syndrome during the reproductive life span. J Clin Endocrinol Metab. 2005;90(8):4650–8. https://doi.org/10.1210/jc.2005-0628.
21. Balaji S, Amadi C, Prasad S, Kasav JB, Upadhyay V, Singh AK, Joshi A. Urban rural comparisons of polycystic ovary syndrome burden among adolescent girls in a hospital setting in India. Biomed Res Int. 2015;2015(1):158951.
22. Kulkarni S, Kulkarni G. Chronic Low dose Step-up Protocol in Ovulation Induction in PCOS. Pract Guide Infertil. 2018;241.
23. Pathak G, Nichter M. Polycystic ovary syndrome in globalizing India: An ecosocial perspective on an emerging lifestyle disease. Soc Sci Med. 2015;146:21–8.
24. Wasir JS, Misra A. The metabolic syndrome in Asian Indians: Impact of nutritional and socio-economic transition in India. Metab Syndr Relat Disord. 2004;2(1):14–23.
25. Kumar G, Dash P, Patnaik J, Pany G. Socioeconomic status scale-modified Kuppuswamy scale for the year 2022. Int J Community Dent. 2022;10(1):1–6.
26. Azziz R, Kintziger K, Li R, Laven J, Morin-Papunen L, Merkin SS, Teede H, Yildiz BO. Recommendations for epidemiologic and phenotypic research in polycystic ovary syndrome: an androgen excess and PCOS society resource. Hum Reprod. 2019;34(11):2254–65. https://doi.org/10.1093/humrep/dez185. (**PMID: 31751476**).
27. Woolcock JG, Critchley HO, Munro MG, Broder MS, Fraser IS. Review of the confusion in current and historical terminology and definitions for disturbances of menstrual bleeding. Fertil Steril. 2008;90(6):2269–80.
28. Ferriman D, Gallwey JD. Clinical assessment of body hair growth in women. J Clin Endocrinol Metab. 1961;21(11):1440–7.
29. Rotterdam ESHRE/ASRM-Sponsored PCOS Consensus Workshop Group. Revised 2003 consensus on diagnostic criteria and long-term health risks related to polycystic ovary syndrome (PCOS). Hum Reprod. 2004;19(1):41–7.
30. Dewailly D, Lujan ME, Carmina E, Cedars MI, Laven J, Norman RJ, Escobar-Morreale HF. Definition and significance of polycystic ovarian morphology: A task force report from the Androgen Excess and Polycystic Ovary Syndrome Society. Hum Reprod Update. 2014;20(3):334–52.
31. Gill H, Tiwari P, Dabadghao P. Prevalence of polycystic ovary syndrome in young women from North India: A community-based study. Indian J Endocrinol Metab. 2012;16(Suppl 2).
32. Vijaya K, Bharatwaj RS. Prevalence and undetected burden of polycystic ovarian syndrome (PCOS) among female medical undergraduate students in South India—a prospective study in Pondicherry. GJRA Glob J Res Anal. 2014;3:63–4.
33. Joseph N, Reddy AG, Joy D, Patel V, Santhosh P, Das S, Reddy SK. Study on the proportion and determinants of polycystic ovarian syndrome among health sciences students in South India. J Nat Sci Biol Med. 2016;7(2):166.
34. Gupta M, Singh D, Toppo M, Priya A, Sethia S, Gupta P. A cross-sectional study of polycystic ovarian syndrome among young women in Bhopal, Central India. Int J Community Med Public Health. 2018;5(1):95–100.
35. Nanjaiah R, Roopadevi V. Prevalence of Polycystic Ovarian Syndrome among Female Students: A Cross-Sectional Study. Natl J Community Med. 2018;9(03):187–91.
36. Jabeen A, Yamini V, Amberina AR, Eshwar MD, Vadakedath S, Begum GS, Kandi V. Polycystic ovarian syndrome: Prevalence, predisposing factors, and awareness among adolescent and young girls of South India. Cureus. 2022. https://doi.org/10.7759/cureus.27943.
37. Shringarpure K, Baxi R, Sheth M, Patel P, Parmar V, Baxi S. Polycystic ovarian syndrome among young women of a university in Central Gujarat – A cross-sectional study. Indian J Public Health. 2023;67(4):575–81. https://doi.org/10.4103/ijph.ijph_1508_22.
38. Deswal R, Narwal V, Dang A, Pundir CS. The prevalence of polycystic ovary syndrome: A brief systematic review. J Hum Reprod Sci. 2020;13(4):261–71.
39. Amiri M, Hatoum S, Buyalos RP, Sheidaei A, Azziz R. The influence of study quality, age, and geographic factors on PCOS prevalence—A systematic review and meta-analysis. J Clin Endocrinol Metab. 2025;dgae917. https://doi.org/10.1210/clinem/dgae917
40. Mishra H, Ali A, Abedin Z. Male selective migration for work: A study of the National Capital Territory (NCT) of Delhi. J Int Womens Stud. 2023;25(1):60–75.
41. Dreier P, Mollenkopf J, Swanstrom T. Place matters: Metropolitics for the twenty-first century. Rev. ed. Univ Press Kansas. 2014.
42. Goyal A, Doomra R, Atkaan N, et al. Is prolonged stress a cause of polycystic ovarian syndrome? A survey from Delhi, National Capital Region. *J Evol Med Dent Sci*. 2021;10(08):505–510. https://doi.org/10.14260/jemds/2021/110
43. Kulkarni SD, Patil AN, Gudi A, Homburg R, Conway GS. Changes in diet composition with urbanization and its effect on the polycystic ovarian syndrome phenotype in a Western Indian population. Fertil Steril. 2019;112(4):758–63.
44. Times of India. Modern Lifestyle Increases PCOS Cases. Times of India. 2012 Jan 20.
45. Pal S. PCOS hitting the young. DNA. 2013 Mar 8.
46. Garari K. PCOS—all you need to know. Asian Age. 2014 Jun 30.
47. United Nations Development Programme. India Factsheet: Economic and Human Development Indicators. 2011.
48. Popkin BM. Nutritional patterns and transitions. Popul Dev Rev. 1993;138–157.
49. Griffiths PL, Bentley ME. The nutrition transition is underway in India. J Nutr. 2001;131(10):2692–700.
50. Nagesh BS. The food and grocery market. Busworld Mark Whitebook. 2013;187–205.
51. Ilyas M, Sumaryani Soemarko D, Yunia Fitria D, Purwito Adi N, Isbayu Putra M, Kualasari Y. Relationship between nutrition intake and the fitness of manufacturing workers in Indonesia. Int J Multidiscip Res (IJFMR). 2021;5(4):1–8.

52. Conlon F. Dining out in Bombay. In: Breckenridge C, editor. Consuming modernity: Public culture in a South Asian world. Univ Minn Press. 1995. p. 90–127.
53. Fernandes L. Restructuring the new middle class in liberalizing India. Comp Stud S Asia Afr Middle East. 2000;20(1):88–104.
54. Misra A, Khurana L. Obesity and the metabolic syndrome in developing countries. J Clin Endocrinol Metab. 2008;93(11_suppl_1)
55. Celermajer DS, Chow CK, Marijon E, Anstey NM, Woo KS. Cardiovascular disease in the developing world: Prevalences, patterns, and the potential of early disease detection. J Am Coll Cardiol. 2012;60(14):1207–16.
56. Chopra SM, Misra A, Gulati S, Gupta R. Overweight, obesity and related non-communicable diseases in Asian Indian girls and women. Eur J Clin Nutr. 2013;67(7):688–96.
57. Khandelwal S, Reddy KS. Eliciting a policy response for the rising epidemic of overweight-obesity in India. Obes Rev. 2013;14(S2):114–25.
58. Ghosh A, Thakur M, Singh SK, Dutta R, Sharma LK, Chandra K, Banerjee D. The Sela macaque (Macaca selai) is a distinct phylogenetic species that evolved from the Arunachal macaque following allopatric speciation. Mol Phylogenet Evol. 2022;174: 107513.
59. Parker J. Pathophysiological effects of contemporary lifestyle on evolutionary-conserved survival mechanisms in polycystic ovary syndrome. Life. 2023;13(4):1056. https://doi.org/10.3390/life13041056.
60. Charifson MA, Trumble BC. Evolutionary origins of polycystic ovary syndrome: An environmental mismatch disorder. Evol Med Public Health. 2019;2019(1):50–63. https://doi.org/10.1093/emph/eoz011.
61. Sharma S, Mishra AJ. Tabooed disease in alienated bodies: A study of women suffering from polycystic ovary syndrome (PCOS). Clin Epidemiol Glob Health. 2018;6(3):130–6.
62. Neel JV. Diabetes mellitus: A "thrifty" genotype rendered detrimental by "progress"? Am J Hum Genet. 1962;14(4):353.
63. Mehreen TS, Ranjani H, Kamalesh R, Ram U, Anjana RM, Mohan V. Prevalence of polycystic ovarian syndrome among adolescents and young women in India. J Diabetol. 2021;12(3):319–25.
64. Mathur A, Tiwari A. Prevalence of polycystic ovary syndromes (PCOS) in adolescent girls and young women: A questionnaire-based study. Indian J Obstet Gynecol Res. 2023;10(3):330–4.
65. Liang SJ, Liou TH, Lin HW, Hsu CS, Tzeng CR, Hsu MI. Obesity is the predominant predictor of impaired glucose tolerance and metabolic disturbance in polycystic ovary syndrome. Acta Obstet Gynecol Scand. 2012;91(10):1167–72.
66. Radha P, Devi RS, Madhavi J. Comparative study of prevalence of polycystic ovarian syndrome in rural and urban population. J Adv Med Dent Sci Res. 2016;4(2):90.
67. Afifi L, Saeed L, Pasch LA, Huddleston HG, Cedars MI, Zane LT, Shinkai K. Association of ethnicity, Fitzpatrick skin type, and hirsutism: A retrospective cross-sectional study of women with polycystic ovarian syndrome. Int J Womens Dermatol. 2017;3(1):37–43.
68. Wolf WM, Wattick RA, Kinkade ON, Olfert MD. Geographical prevalence of polycystic ovary syndrome as determined by region and race/ethnicity. Int J Environ Res Public Health. 2018;15(11):2589.
69. Sendur SN, Yildiz BO. Influence of ethnicity on different aspects of polycystic ovary syndrome: A systematic review. Reprod Biomed Online. 2021;42(4):799–818.
70. Khubchandani J, Soni A, Fahey N, Raithatha N, Prabhakaran A, Byatt N, Allison JJ. Caste matters: Perceived discrimination among women in rural India. Arch Womens Ment Health. 2018;21:163–70.
71. Shaikh M, Miraldo M, Renner AT. Waiting time at health facilities and social class: Evidence from the Indian caste system. PLoS One. 2018;13(10)
72. Khalfe S, Rafi U, Fatima B. Prevalence and awareness of polycystic ovarian syndrome among medical students of Karachi, Pakistan. J Shifa Tameer-E-Millat Univ. 2024;6(2):78–82. https://doi.org/10.32593/jstmu/Vol6.Iss2.260
73. Kamrul-Hasan ABM, Aalpona FTZ, Mustari M, Selim S. Prevalence and characteristics of women with polycystic ovary syndrome in Bangladesh–A narrative review. Bangladesh J Endocrinol Metab. 2023;2(1):20–8.
74. Quadir F, Barua M, Pathan F, Kuryshi SA, Chakma PJ, Barua B, Islam M. Frequency of Polycystic Ovary Syndrome among the Students of a Medical College in Dhaka City. Int J Diabetes Metab Disord. 2020;5(3):91–7.
75. Sangraula H, Paudel KR, Sharma M. Metformin and troglitazone in the treatment of female infertility associated with polycystic ovarian syndrome. JNMA J Nepal Med Assoc. 2009;48(176):335–9.
76. Moghul SN. 1 in 5 women affected by PCOS in India! But fret not, we have the solution. India Times. 2015 Sept 7.
77. Rajashekarappa MR. Recent changes in the Hindu joint family: An evolution towards modernity. Int J Multidiscip Res IJFMR. 2023;5(4):1–10.
78. Pilisuk M, Parks SH. The place of network analysis in the study of supportive social associations. Basic Appl Soc Psychol. 1981;2(2):121–35.
79. Klever P. Multigenerational stress and nuclear family functioning. Contemp Fam Ther. 2005;27(3):233–50.
80. Ambrosini GL, Oddy WH, Robinson M, O'Sullivan TA, Hands BP, De Klerk NH, Silburn SR, et al. Adolescent dietary patterns are associated with lifestyle and family psycho-social factors. Public Health Nutr. 2009;12(10):1807–15.
81. Kaiser Foundation Health Plan and Hospitals. Kaiser Foundation Health Plan and Hospitals report 2023 financial results. Kaiser Permanente. 2023 Aug 4. https://about.kaiserpermanente.org/news/press-release-archive/kaiser-foundation-health-plan-and-hospitals-report-2023-financial-results
82. Rea A. How serious is America's literacy problem? Library J. 2020 Apr 29. https://www.libraryjournal.com/story/How-Serious-Is-Americas-Literacy-Problem
83. Merkin SS, Azziz R, Seeman T, Calderon-Margalit R, Daviglus M, Kiefe C, Matthews K, Sternfeld B, Siscovick D. Socioeconomic status and polycystic ovary syndrome. J Womens Health (Larchmt). 2011;20(3):413–9. https://doi.org/10.1089/jwh.2010.2303.

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